LAB EXPERIMENT

ITA0506-COMPUTER VISION

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EXPNO:7

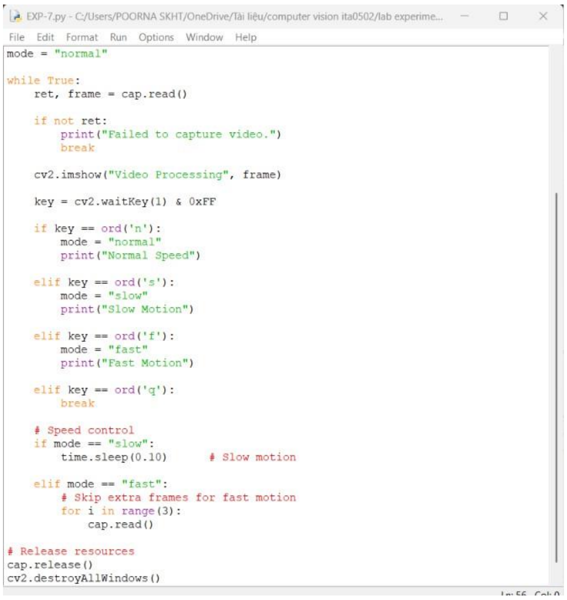
Aim:

To capture live video using a webcam and display it in normal speed, slow motion, and fast motion using Python and OpenCV.

Algorithm:

1. Import the required libraries (cv2 and time).
2. Open the webcam using cv2.VideoCapture(0).
3. Check if the webcam is opened successfully.
4. Capture video frames continuously using a loop.
5. Display the captured video frame.
6. Press 'N' to view the video at normal speed.
7. Press 'S' to display the video in slow motion by adding a delay.
8. Press 'F' to display the video in fast motion by skipping frames.
9. Press 'Q' to stop the video and exit.
10. Release the webcam and close all OpenCV windows

Code:



EXP-7.py - C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experime...

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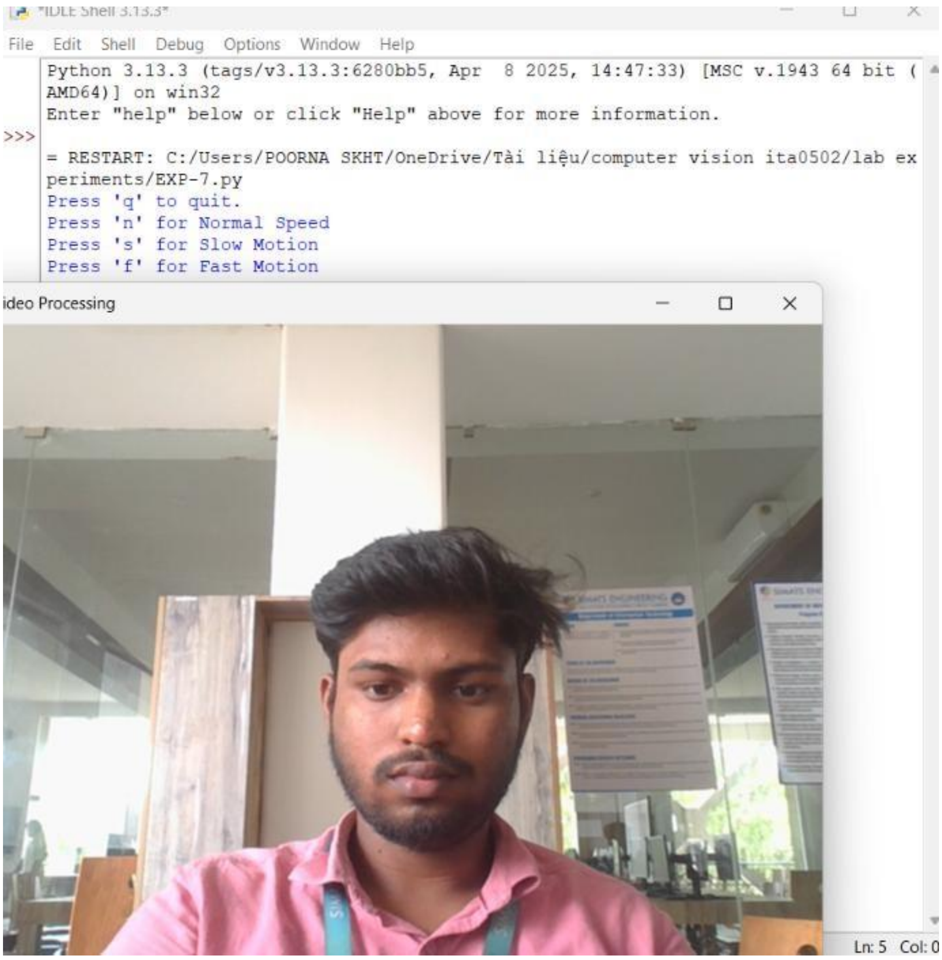
```python
mode = "normal"

while True:
    ret, frame = cap.read()

    if not ret:
        print("Failed to capture video.")
        break

    cv2.imshow("Video Processing", frame)

    key = cv2.waitKey(1) & 0xFF

    if key == ord('n'):
        mode = "normal"
        print("Normal Speed")

    elif key == ord('s'):
        mode = "slow"
        print("Slow Motion")

    elif key == ord('f'):
        mode = "fast"
        print("Fast Motion")

    elif key == ord('q'):
        break

    # Speed control
    if mode == "slow":
        time.sleep(0.10)      # Slow motion

    elif mode == "fast":
        # Skip extra frames for fast motion
        for i in range(3):
            cap.read()

# Release resources
cap.release()
cv2.destroyAllWindows()
```

OUTPUT :



RESULT: The live video was successfully captured and displayed in **normal speed**, **slow motion**, and **fast motion** using Python and OpenCV.